

Agile and Wellbeing — Stress, Empowerment, and Performance in Scrum and Kanban Teams

Maarit Laanti
Nokia
Maarit.Laanti@iki.fi

Abstract

Little research data exist about agile teams and wellbeing. After changing our software engineering mode to agile, we wanted to find out if people experienced more or less stress than before. This study is based on a company-wide survey of 466 software engineering practitioners. We asked about their subjective feelings about stress, empowerment, and performance in their respective engineering teams after they had started to use agile methods. The results reveal that the feelings of higher performance improvement and sustainable pace are related, and that this difference is statistically significant. Respondents who feel that their team is empowered also feel less stress. However, no significant difference in feelings of stress and empowerment was found between respondents working in Kanban and Scrum mode after transition to agile. The group that was improving its performance because of agile was also reporting a better workload balance. The group that was experiencing worse performance because of agile methods was also more stressed. The differences may be explained by the management styles being practiced in the teams. The results are important as those verify in industrial setting what some agile books suggest: limiting workload and team empowerment has an impact on performance and stress. This contributes to our understanding how agile teams should be managed.

methods have been widely recognized in the software industry [3, 4]. In addition, the existing numerous publications on agile methods on both scientific and non-scientific forums mainly report positive impacts of agile adoption. However, adoption of agile methods has proven to be a challenging task [5]. It has been claimed that agile adopters are often unaware of what agile adoption really means, and how broad a change is actually required [2].

Agile methods are reported to bring business value to users but only little research exists about whether this is actually happening at the expense of the wellbeing of the personnel in agile projects. Syed-Abdullah, Holcombe, and Gheorge [6] have studied wellbeing in teams using Extreme Programming while Whitworth and Biddle have studied the social nature of agile teams [7].

It is being said that Scrum, for example, resembles a typical project crisis mode: all people work just on the most important tasks at hand; priorities are made very explicit and there is pressure from the deadline.

Traditionally, when the crisis is over people will return back to their normal working conditions and maybe need a break (or less busy hours) to recover from the overwork. Because agile teams are supposed to be capable of working with a constant flow (and constant pace) one of the twelve agile principles is the principle of **sustainable pace**: Agile processes promote sustainable development. Product managers and owners, development teams, and people responsible for testing should be able to maintain a constant pace indefinitely [8]. So the central question is: how well can agile teams adapt to constant pressure of biweekly sprint deadlines and implementation priorities?

In this research we study wellbeing in agile teams focusing on performance, empowerment, and stress level. The data are based on a survey done in December 2010 with 466 respondents from various teams in a large embedded software development organization at Nokia. The main agile model in use was scaled-up Scrum [8] as defined in [9] and [10] but some teams were also using Kanban [11] or a

1. Introduction and Related Literature

1.1. Sustainable Pace

According to a Forrester study, agile has reached mainstream: in their survey the majority of respondents stated that agile most closely reflects their development methods [1]. The growth of the usage of agile methods in last three years (from 17% up to 35%) has been remarkable [2]. This is likely due to same reason why agile methods have gained popularity in Nokia; that the benefits of agile

mixture of Scrum and Kanban. At the time of the research, Kanban was used only at team level, and mainly for maintenance. Thus in this paper Kanban is referred to as an agile method like in [11] though Kanban is rather based on lean than agile principles [12].

This paper provides a reference point for similar wellbeing studies on other product development practices. The results indicate that for those people whose feelings of stress are connected to working method they use, team empowerment is connected to having less stress, and that sustainable pace and teams' performance are related.

1.2 Background

In Scrum, sustainable pace is linked with empowerment [13] and maintained by measuring the team velocity. Effort is planned to the next timebox, a.k.a. sprint, based on the achievement from the previous timebox. Empowered teams measure their velocity, i.e. how many story points they can complete on the average in one sprint. A story point is an abstract measurement unit for relative work amount size estimation. Each user story is estimated separately in terms of story points [14]. An empowered Scrum team plans only to do as much work as they completed in the previous sprint and is thus able to work following a sustainable pace. This is opposite to traditional plan-driven development, where management is often setting deadlines and teams are cutting quality and working overtime to get the release out [13].

Empowered Scrum teams are also able to make technical decisions by themselves, whereas business decisions are the responsibility of product manager a.k.a. Product Owner. "Product Owner" is a Scrum role responsible for defining and communicating the vision for the product being built and scoping the content when necessary. Agile development is based on the discovery that traditional software products have many features that are not at all or seldom used [15]. Thus the profitability of agile methods is also based on focusing on doing the right things with very good quality rather than implementing a broader set of features whereas traditional methods in their extreme try to push all planned features out of the development pipeline at the same time. This often results in delays and unclarity about the final release date since the last weeks of the product development project are spent in testing and improving the quality to an acceptable level.

The fear of empowerment that traditional project management seems to have is related to the fear of breaking commitment to a delivery at a final project

delivery deadline. Yet in Scrum teams there is willingness to achieve a release after every sprint. Peer pressure from agile team members is replacing the former management command-and-control type of target setting. This visibility may cause significantly higher stress for some individuals. From the data available, we can only guess the reasons. Lack of competence, personality type, and not liking teamwork are some of the suggested reasons. Agile is demanding a lot of openness and teamwork but the fact is that we really do not know.

2. Research Approach

In this section, the research aims, context, and data collection and analysis methods are presented. The assumption is that the studied variables have complex and unknown relations with each other. Thus the research done is exploratory research. This is why there are several research questions in this paper.

2.1 Research Aims

Several wellbeing studies have been conducted in traditional product development contexts, but only very few have been done within an agile setting. The aim of this research was to test the most typical claims in agile literature regarding subjective wellbeing in agile teams.

Subjective feelings are important because "whatever we perceive is organized into patterns for which we the perceivers are largely responsible" [16]. Kurtz and Snowden [17] propose that for those parts that we do not understand, "we fill the gaps to create an experience-based pattern on which we act." These experiences, in turn, are visible only by means of inquiring about subjective feelings. Thus, it can be argued that they represent, when asked from a significant portion of the population, the reality as it exists in the studied organization. The consideration of whether this reality is transferable to another organization falls outside the scope of this study. We maintain those subjective feelings and their subsequent correlations (or non-correlations) with the determinant factors that can act as a vehicle for discovering patterns of behaviour, which inherently are present in an agile organization. We acknowledge that the patterns may repeat for a time, but we cannot guarantee and foresee that they would continue to repeat [16].

One of the typical statements with Scrum is that performance can increase even though teams are

working with sustainable pace. This seems to be a typical discussion topic in blogs and other forums where agile practitioners discuss with each other.¹ These forums provide rich material for researchers to study different opinions of agile practitioners.

Working overtime is one of the traditional management methods used, so the Scrum statement that sustainable pace increases performance is hard for seasoned project managers to believe. Thus, we set the first research question as:

Q1. Has performance increased in all teams alike? Has sustainable pace anything to do with performance increase?

One of the elements that Scrum and sprint planning advocate is “stress-free productivity.” In a survey like this we cannot measure productivity objectively, but the perceived (self-measured) performance. This perceived performance can then be compared to a perceived feeling of how empowered a team is and what the team stress level is.

Not all people agree with the idea of “stress-free productivity”, though. Traditional project management is based on thinking that more can be achieved if the team over-commits rather than under-commits.² We set the second research question as:

Q2. Do feelings of stress and team empowerment relate?

Here we study the claim that the work mode and feelings of stress and empowerment are related to each other. We study if the feelings are different for people working with different methods (Kanban or Scrum). Ken Schwaber has been claiming in his blog that “Kanban is just a death march without a pause” [18]. If this is true, then Kanban teams should have more stress and feel less empowered. Thus we set the third and fourth research questions as:

Q3. Do Kanban and Scrum teams feel similar stress, compared to traditional development?

Q4. Do Kanban and Scrum teams feel equally empowered, compared to traditional development?

2.2 Research Context

The survey was sent semi-randomly to different parts of the software engineering organization. We tried to send the survey to over 10,000 people but our survey tool could not handle this load and kept reporting problems. As a result, we do not know the actual number of sent invitations. Eventually, we received 466 unique answers. The answers were anonymized for the analysis. There would have been an option to repeat the survey with a better tool but when we analyzed the 466 answers we concluded that they were representative enough as the responses were spread over many roles and different types of teams.

Most of the respondents were working in Scrum teams. Respondents had the following roles:

- 59% Scrum team members,
- 22% Scrum Masters,
- 10% Product Owners,
- 4% Coach,
- 23% Other roles.

Selection of multiple roles was possible, so the sum exceeds 100%.

The teams were also asked if they were able to use all Scrum practices or if they were able to use just some. Following only a subset of practices from Scrum is called ScrumBUT according to Nokia Test³, and several teams were using this approach [19]. Some teams were also using a combination of Scrum and Kanban, called Scrumban.

Respondents were using different methods:

- 39% said they are following Scrum
- 37% were following ScrumBUT
- 10% were following Scrumban
- 5% were following Kanban
- 5% were following traditional methods
- 3% said they were following some other method, and
- 2% said this question is non-applicable to them.

3. Data Collection and Initial Results

This research was conducted as a questionnaire-based survey. The questionnaire was open for two and a half weeks in the end of December 2010. In the survey we also asked for opinions on Kanban, the existing backlog tool, and what kind of tool the Kanban users would prefer. In this study, however,

¹ A related InfoQ article listing many links to agile blogs and other forums discussing overtime can be found here <http://www.infoq.com/news/2008/05/sustainable-pace>

² See a related blog post, e.g. in <http://scrumftw.blogspot.com/2011/02/overcommitment-is-better-than.html>

³ Nokia Test consists of several questions that test if a team is following all Scrum rules or not. The test was developed in Nokia Siemens Networks.

we limit ourselves to wellbeing and work mode questions.

The work mode questions were multi-choice questions with both positive and negative options. Overall, the feedback towards agile methods was very positive:

- When people were asked about the workload with seven answer options, the biggest respondent group, i.e. 29% felt they were working with sustainable pace.
- The majority (64%) of respondents felt their performance had increased.
- Of those who stated that changing to use agile methods had impacted their stress level, a slightly larger group — 12% larger — stated that the impact had been positive.
- When simply asked if the respondents were working with sustainable pace with only Yes or No options, a clear majority (82%) felt their team was working with sustainable pace. The same is true also with empowerment as almost as many (71%) felt their team was empowered.
- 55% of respondents would like to stay working in agile way, and only 12% would like to go back to traditional way of working.

All response options to the work mode questions and the distribution of answers were as follows:

Q1. How is your team's workload after taking agile into use?

- 14% – There is more work, but it is not because of agile,
- 12% – There is more work / more hectic because of agile,
- 29% – There is workload balance / sustainable pace,
- 7% – There is less work because better communication helps us optimize the solutions & content; we also do scope management,
- 4% – There is simply too much work,
- 25% – Agile has no impact to workload,
- 10% – Other.

Q2. How is your team's performance after taking agile into use?

- 3% – Performance has decreased significantly,
- 8% – Performance has decreased,
- 38% – Performance has increased,
- 3% – Performance has increased significantly,
- 26% – Performance has improved and it is still improving,

- 26% – Other.

Q3. How is your stress level after moving to agile?

- 27% – Better because of agile,
- 34% – The same,
- 15% – Worse because of agile,
- 23% – Not related to agile methods.

Q4. Is your team “empowered”?

- 71% – Yes,
- 29% – No.

Q5. Does your team work at “sustainable pace”?

- 82% – Yes,
- 18% – No.

Q6. Would you go back to the old way of working?

- 55% – No,
- 31% – I do not know,
- 12% – Yes,
- 2% – I do not know as I have never worked with non-agile methods.

4. Analysis of Results

The quantitative analysis done is based on the Kruskal-Wallis H test [20, 21]. This test is a nonparametric test that does not assume normality. It tests the equality of population medians among the groups. Instead of using the variance, it is replacing the actual values with ranks — otherwise it is identical to the one-way ANOVA (analysis of variance) parametric test. The Kruskal-Wallis H test is like the Mann-Whitney U test, but it can be used with multiple values where the Mann-Whitney U is limited to nominal variables with only two values.

It is assumed in the Kruskal-Wallis test that the observations in the data set are independent of each other and that the observations must be drawn from the population by the process of random sampling.

4.1. Kruskal-Wallis H Test for Sustainable Pace and Performance

The relation of sustainable pace and performance increase was studied with the Kruskal-Wallis H test. With the Kruskal-Wallis H test the mean ranks of samples from the populations are expected to be the same, i.e.:

$$H_0: \mu_1 = \mu_2 = \mu_3$$

$$H_A: \mu_i \neq \mu_j$$

For this test we are omitting the 26% group who answered “Other” (i.e. with free text answer) — most of them stated that their performance is not related to agile methods.

Figure 4.1 shows how the opinions distribution of how agile impacts performance differs depending if the respondent is working in sustainable or non-sustainable pace though the median of both groups is the same.

The mean ranks in Table 4.1 indicate that teams that were working with sustainable pace had increased their performance more than those who did not work with sustainable pace.

From Table 4.2 we can see that the means of the two groups are unequal and that the mean of the group that was working with sustainable pace is higher (so their performance had increased more). This result is significant with ($H= 6.208$, 1 d.f.⁴, $P < 0.05$).

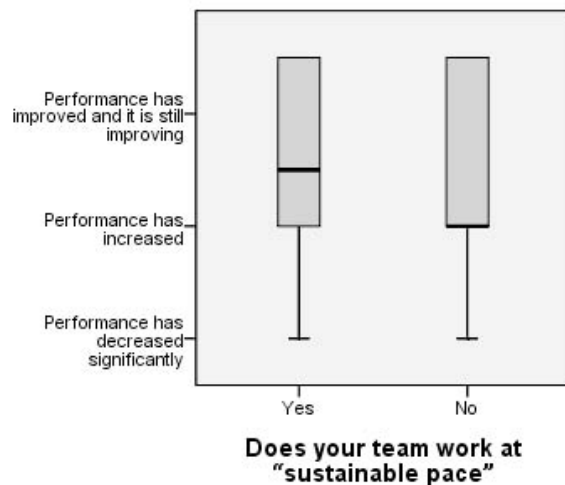


Figure 4.1 A boxplot figure of performance opinions and whether or not the team works with sustainable pace.

Table 4.1 Mean ranks of performance and sustainable pace.

Team's performance after taking agile into use		
	N	Mean rank
Sustainable pace		
Yes	254	159.44
No	53	127.95
Total	307	

Table 4.2 Kruskal-Wallis test statistics, grouping variable: Sustainable pace.

Team's performance after taking

⁴ d.f. = degree of freedom

agile into use	
Chi-Square	6.208
df	1
Asymp. Sig.	0.013

4.2 Kruskal-Wallis H Test for Stress and Empowerment

Table 4.3 Mean ranks of stress and team empowerment.

Stress level after moving to agile		
	N	Mean rank
Empowered		
Yes	262	161.85
No	95	226.30
Total	357	

Table 4.4 Kruskal-Wallis test statistics, grouping variable: empowerment.

Stress level after moving to agile	
Chi-Square	31.714
Df	1
Asymp. Sig.	0.000

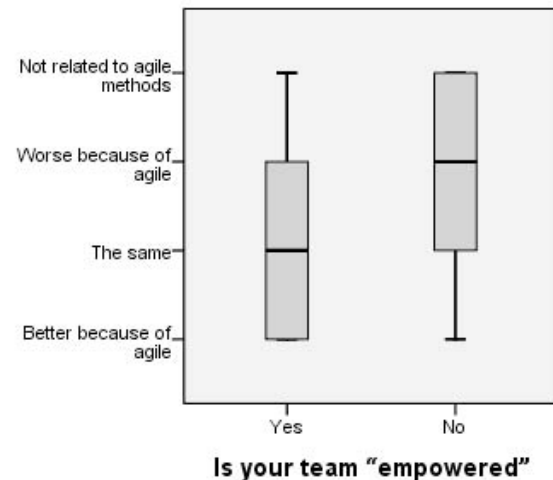


Figure 4.2 Boxplot figure of opinions of stress level opinions and team is empowerment.

The research question that stress and empowerment are not connected was tested with the Kruskal-Wallis H test. For this test we are omitting the 23% group who answered “Not related to agile methods” as in Kruskal-Wallis H Test the survey answers need to be ordered.⁵

⁵ However, the survey response “Not related to agile methods” has numerical value of 4, i.e. it is the biggest of all response options. “Worse because of agile” has 3, “The same” has 2 and “Better because of agile” has 1. When it is included in Kruskal-Wallis H Test, the result is significant with ($H= 27.301$, 1 d.f., $P < 0.01$).

Figure 4.2 shows how the median and distribution of stress levels differ between empowered and non-empowered teams.

The mean ranks in Table 4.3 indicate that empowered teams are more likely to have less stress because of agile. Table 4.4 shows that the differences are significant.

From the results we can see that the means of the two groups are unequal and that the empowered teams are more likely to have less stress. This result is significant with ($H= 27.301$, 1 d.f., $P < 0.01$).

4.3 Kruskal-Wallis H Test for Stress in Kanban and Scrum Teams

The question if feelings of stress were similar between Kanban and Scrum teams was tested with the Kruskal-Wallis H test.

Again, we omitted the group of 23% who think that stress and agile methods are not related. The mean ranks in Table 4.5 indicate that Kanban teams might have slightly less stress, but Table 4.6 shows that these differences are not significant.

Table 4.5 Mean ranks of stress levels in different teams.

Teams way of working	Stress level after moving to agile	
	N	Mean rank
Scrum	144	162.07
ScrumBUT	136	171.85
Scrumban	37	183.69
Kanban	16	133.94
Total	333	

Table 4.6 Kruskal-Wallis test statistics, grouping variable: team's way of working.

Stress level after moving to agile	
Chi-Square	4.197
df	3
Asymp. Sig.	0.241

The results indicate that no significant difference was found in respondent's feeling of stress based on which agile method they used.⁶

4.4 Kruskal-Wallis H Test for Empowerment in Kanban and Scrum Teams

The research question if Kanban and Scrum teams felt equally empowered was tested with the Kruskal-

Wallis H test. Figure 4.3 shows the empowerment and work mode.

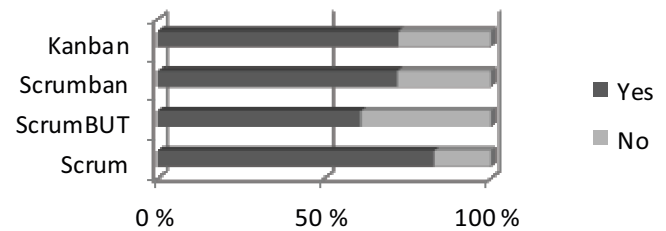


Figure 4.3. Team Empowerment (Yes / No) by work mode (Kanban / Scrumban / ScrumBUT / Scrum).

The mean ranks in Table 4.7 indicate that Scrum teams are likely to feel more empowered than Kanban teams, but table 4.8 shows that this difference is not significant.

Table 4.7 Mean ranks of empowerment in different teams.

Team's way of working	Empowered	
	N	Mean rank
Scrum	181	100.88
Kanban	22	111.18
Total	203	

The results indicate that no significant difference was found between respondents' feeling of empowerment depending on their use of Scrum or Kanban after transition to agile methods.⁷

Table 4.8 Kruskal-Wallis test statistics, grouping variable: team's way of working.

Empowered	
Chi-Square	1.348
df	1
Asymp. Sig.	0.246

5. Discussion

5.1 How are Over-commitment, Stress, and Performance Related?

In the absence of more objective measures of empowered teams and productivity, we can see how the reported stress levels compare to subjective opinions of teams' performance. Figure 5.1 shows these data. From the graph we can see that

⁶ The result would be the same with Asymp. Sig. 0.766, even if including also the group that thinks stress and agile methods are not related.

⁷ This test was also repeated with Scrum and ScrumBUT teams, and it was discovered that this difference was significant — Scrum teams felt more empowered ($H= 21.571$, 1 d.f., $P < 0.01$).

respondents who said that their stress level is the same or better are also mostly those who also stated their teams' performance has increased or is still

improving. These are people who seem to benefit from working in agile mode.

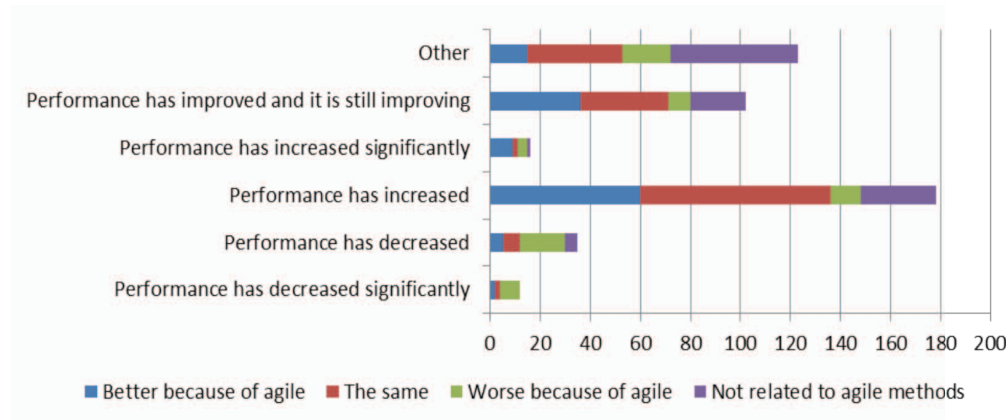


Figure 5.1 Stress level opinions grouped by performance.

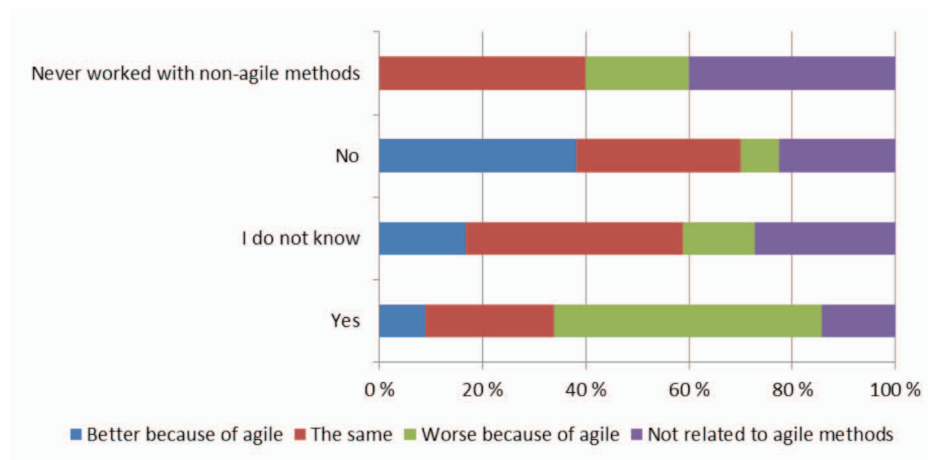


Figure 5.2 Stress level opinions grouped by if respondent would like to go back to traditional methods.

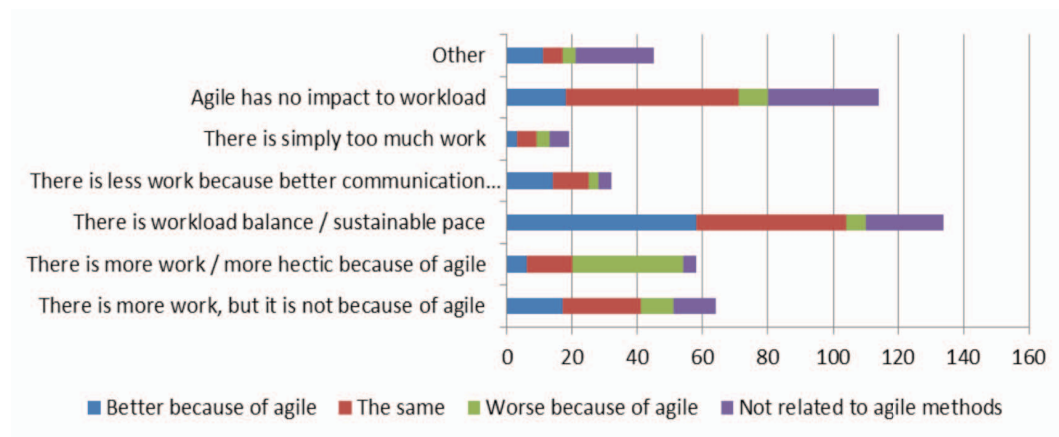


Figure 5.3 Stress level opinions grouped by workload opinions.

Table 6.1 Summary of Research questions with results.

Research question	
Q1. Has performance increased in all teams alike? Has sustainable pace anything to do with performance increase?	Not agreed. Sustainable pace leads to performance increase. Notice that there is a group of people who think agile is not impacting performance.
Q2. Do feelings of stress and team empowerment relate?	Not agreed. Empowered teams have less stress.
Q3. Do Kanban and Scrum teams feel similar stress, compared to traditional development?	Agreed. There was no statistical difference found in how much stress Kanban and Scrum teams felt compared to time before agile change.
Q4. Do Kanban and Scrum teams feel equally empowered, compared to traditional development?	Agreed. There was no statistical difference found in how empowered Kanban and Scrum teams felt compared to time before agile change.

It would be worth of studying what leads to high performance with less stress. Scrum suggests continuous improvement as one answer. It could also be that these teams work with better tools than other teams. E.g. Kokko [22] has suggested that a well-working and jointly maintained Continuous Integration system improves the team's and project's productivity and speed of execution.

There were 5 respondents who stated that their performance has decreased and their stress level was worse because of agile. These were the ones who were clearly unhappy with agile methods. It would be interesting to know why.

Figure 5.2 reveals that people who felt stressed because of agile methods are also over-represented among those who would have liked to go back to the old way of working.

Figure 5.3 shows that these feelings of stress are highly linked with the respondents' workload: there seem to be teams that were able to work with sustainable pace and others that had not made this transition — i.e. they had higher stress and higher workload because of agile.

Combining the result from Section 4.1 — teams who were working with sustainable pace also had higher performance — it can be concluded that over-commitment will not lead to better results.

There could be an alternative to over-commitment, high stress, and high workload. A different management style, which builds on empowered teams and working smarter, not harder, with high focus might lead to better results. Management theories exist about how to lead empowered organizations — our study suggests that some managers relying on traditional management methods might benefit from such training.

5.2 Limitations of the Study

The biggest limitation of this study is that it is purely survey-based and the real performance of teams was not measured. A survey like this cannot

reveal other contributing factors outside the survey instrument that could have revealed in in-depth interviews.

We also limit ourselves on the studied factors only. Other factors that may have impact on the result, such as the length of the agile experience or the organization level, are excluded from the study.

The survey was done in one company only, so it might have been biased by the organization culture and the specific interpretation of agility. This raises the question regarding external validity (i.e. how largely these results can be generalized).

This study was covering several organizational units in Nokia and various types of projects. Thus the expectation is that there is variation of project and sub-organization cultures and how agile has been interpreted and deployed. This suggests that the results are more general, rather than specific to this one company. However, it is suggested that different companies and organizations would conduct related studies in order to validate our assumptions and the generality of these findings.

It is typical to measure stress with many latent variables. E.g. the Siegrist model uses two summary measures of work stress: imbalance, the ratio between extrinsic effort (demands on the job) and rewards (money, esteem, and status control), and over-commitment, a psychological coping style associated with the inability to withdraw from work obligations [21]. Measuring stress with a scale from high to low before and after agile would have given more room for interpretations than a single question.

The Job Demands – Job Decision Latitude Model [24] is providing the scientific basis for studies on psychological stressors such as working long and hard compared to worker's authority on making decisions on the job and variety of skills used by worker on the job. However, there are studies showing that subjective appraisals of work stress correlate more strongly with self-reported depression than objective studies of work conditions do [25]. Even though a more thorough study would be

required to capture different perceptions regarding latent variables of stress, the perception cannot be totally neglected either. The fundamental principle of sociology, known as the Thomas theorem, states that "It is not important whether or not the interpretation is correct – if men define situations as real, they are real in their consequences" [26].

The Kruskal-Wallis H test measures the differences of medians among the groups. This study detected no differences in stress or empowerment variables between groups using Scrum and Kanban. This, however, does not mean that Scrum and Kanban teams are the same.⁸ On the contrary, it is suggested that further studies are conducted to study differences directly between Scrum and Kanban teams.

Also, because of the problems with the survey tool the response percentage remains unknown.

6. Conclusions

In this study, four research questions related to wellbeing in agile teams were studied. Traditionally, wellbeing in software development teams has been seen as important but not as important as getting the products ready. Based on the results of our survey this kind of dualism may not have scientific support. In fact, it may be plausible that agile teams with high wellbeing actually have better performance. This is in alignment with previous studies of empowerment and performance [27].

The results are important as those verify in industrial setting what some agile books suggest: limiting workload and team empowerment has an impact on performance and stress. This contributes to our understanding how agile teams should be managed.

The four research questions were studied using quantitative methods. First, we investigated if a team's sustainable pace leads to better performance. This was supported by our survey results. Second, we discovered that empowered teams cope better with stress. Finally, no statistical difference was found in stress and empowerment between Scrum and Kanban teams. Table 6.1 presents a summary of the research questions with the corresponding findings.

The finding that empowered teams cope better with stress was no surprise and is aligned what psychologists have for a long time stated for

individuals: that feelings of empowerment lead to better control of stress [27]. However, this to our knowledge has not been previously verified in software development and with Scrum teams.

It is notable that even when we asked about team empowerment, every person answered this survey individually, so people in the same team may have answered this question differently as they might have disagreed about their team being empowered or not. This could open further possibilities for research comparing teams instead of asking from individuals, or comparing the opinions of individuals within the same team.

The results regarding sustainable pace and better performance need further verification by real performance metrics done at the team level. One could always argue that subjective means like this survey do not reveal the real situation. Yet the high number of respondents (466) would make it unlikely that people would just reply to the survey to please the researcher. We believe that it is far more plausible that people have answered this survey because what they have experienced in their everyday life. This means that organizations should take the transition as a serious challenge, and find means such as agile coaching in order to help teams and managers to transform from over-committing, stressful, and un-performing working mode to sustainable, less-stressful, and better-performing mode.

Last but not least, special attention needs to be given to the small population that felt worse because of agile and whose performance had also decreased. Is this because of a problem in the existing way how agile is deployed that could be corrected? What made them favor the traditional way of working? Is there anything we could have changed in the existing setting to make them feel better?

Acknowledgements

The author would like to thank Professor Pekka Abrahamsson for his support and guidance.

References

- [1] West D, Grant T, Agile Development, Mainstream Adoption Has Changed Agility, Forrester Research, 2010.
- [2] Schwaber K, Laganza G, D'Silva D, The truth about agile processes: frank answers to frequently asked questions, Forrester Report, 2007.

⁸ Here, it should be noted that the questionnaire emphasis was set to "after taking agile into use". A study comparing Scrum mode directly to Kanban mode might give different results, and is suggested.

- [3] Laanti M, Salo O, Abrahamsson P, Agile methods rapidly replacing traditional methods at Nokia: A survey of opinions on agile transformation. *Information & Software Technology* 53(3): 276-290 (2011)
- [4] Laanti M, Agile Transformation Study at Nokia - One Year After. *LESS* 2010: 3-19
- [5] Svensson H, Höst M, Introducing an agile process in a software maintenance and evolution organization. *Proceedings of 9th European conference of maintenance and reengineering*, 2005.
- [6] Syed-Abdullah S, Holcombe M, Gheorge M, The Impact of an Agile Methodology on the Well Being of Development Teams. *Empir Software Eng* (2006) 11: 143–167, DOI 10.1007/s10664-006-5968-5.
- [7] Whitworth E, Biddle R, The Social Nature of Agile Teams. *Proceeding of AGILE 2007*. ISBN: 0-7695-2872-4.
- [8] Schwaber K, Beedle M, *Agile Software Development with Scrum*, Prentice-Hall, Inc., 2002.
- [9] Laanti M, Implementing program model with agile principles in a large software development organization. In: *Proceedings of 32nd annual IEEE international conference on Computer software and applications (Compsac '08)*, 2008.
- [10] Leffingwell D, Aalto J-M, *A Lean and Scalable Information Model for the Agile Enterprise*. Whitepaper, 2009.
- [11] Kniberg H, Skarin M, *Kanban and Scrum—making the most of both*. Lulu.com, 2010. ISBN-10: 0557138329
- [12] Ikonen M, *Lean Thinking in Software Development : Impacts of Kanban on Projects*. Dissertation, University of Helsinki. ISBN: 978-952-10-7410-3
- [13] Schwaber K, *Enterprise and Scrum*, Microsoft Press 2007. ISBN-10: 0735623376, ISBN-13: 978-0735623378.
- [14] Cohn M, *Agile Estimating and Planning*. Prentice Hall, 2005. ISBN-10: 0131479415. ISBN-13: 978-0131479418.
- [15] Larman C, *Agile & Iterative Development. A Manager's Guide*. Addison-Wesley Professional 2003. ISBN-10: 0-13-111155-8.
- [16] Douglas M, *Purity and Danger, An Analysis of Concepts of Pollution and Taboo*, 1966. Routledge and Keegan Paul, ISBN: 0710012993
- [17] Kurtz, C.F, Snowden, D, The new dynamics of strategy: Sense-making in a complex and complicated world, *IBM Systems Journal*, 42(3): 462-483, 2003.
- [18] Ken Schwaber's Blog, Telling It Like It Is Waterfall, Lean/Kanban, and Scrum
<http://kenschwaber.wordpress.com/2010/06/10/waterfall-leankanban-and-scrum-2/>
- [19] Santana C, Caetano D, Alexandre S, and Rocha V, Measuring the Effectiveness of Nokia Test in Very Small Teams. *Proceedings of Software Engineering and Applications (SEA) 2009*
- [20] Field A, *Discovering Statistics using SPSS*, 3rd edition. Sage Publications Ltd, 2009. ISBN 978-1-84787-906-6.
- [21] *SPSS Base 16.0 User's Guide*. SPSS Inc. 2007. ISBN-10: 0-13-603600-7, ISBN-13: 978-0-13-603600-5.
- [22] Kokko H, *Large Scale Continuous Integration*. Scan Agile conference 2011.
- [23] Siegrist J, Peter R, Job stressors and coping characteristics in work related disease: issues of validity. *Work & Stress*. 1994;8:130 –140.
- [24] Ganster D, Schaubroeck J, Work Stress and Employee Health. *Journal of Management* 1991; 17; 235. DOI: 10.1177/014920639101700202
- [25] Tennant C, Work-related stress and depressive disorders. *Journal of Psychosomatic Research* 51 (2001) 697– 704.
- [26] Merton R, The self-fulfilling prophecy. *The Antioch Review* Vol. 8, No. 2 (Summer, 1948), pp. 193-210.
- [27] Erstad M, Empowerment and organizational change. *International Journal of Contemporary Hospitality Management* 9/7, 1997, p. 325-333. ISSN 0959-6119.